Question Number : 274 Question Id : 6406531290047 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
The Cobb-Douglas production function $F(K,L)=K^{p}L^{q}$
Options :
6406534343465. ✖ Never exhibits decreasing returns to scale
6406534343466. ✔ Shows constant returns to scale if p+q=1
6406534343467. ✖ Has constant marginal product of labour
6406534343468. ✖ Has constant marginal product of capital

Question Number : 275 Question Id : 6406531290048 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
When average product is decreasing in labor, marginal product is greater than average product.
Options :
6406534343469. ✖ TRUE
6406534343470. ✔ FALSE

Question Number : 276 Question Id : 6406531290049 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
When average product is increasing in labor, marginal product is less than average product.
Options :
6406534343471. ✖ TRUE
6406534343472. ✔ FALSE

Question Number : 277 Question Id : 6406531290050 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
When average product $\mathbf{AP}_L$ is at a maximum, then marginal product is equal to average product
Options :
6406534343473. ✔ TRUE
6406534343474. ✖ FALSE

Game Theory and Strategy

Section Id :	64065392201
Section Number :	14
Section type :	Online
Mandatory or Optional :	Mandatory

Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653204541
Question Shuffling Allowed :	No

Question Number : 278 Question Id : 6406531290051 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : GAME THEORY AND STRATEGY (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534343475. ✔ YES

6406534343476. ✖ NO

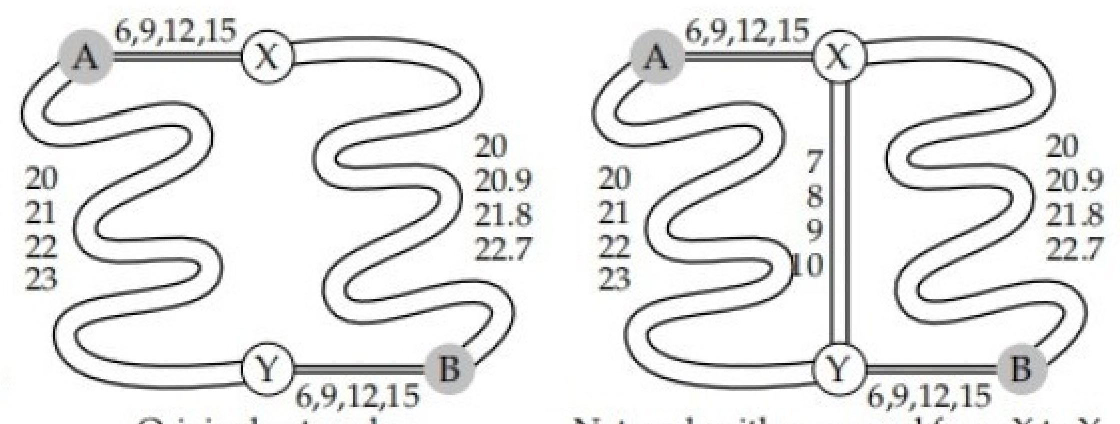
Sub-Section Number :	2
Sub-Section Id :	640653204542
Question Shuffling Allowed :	No

Question Id : 6406531290052 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (279 to 280)

Question Label : Comprehension

(Choosing a route) Four people must drive from A to B at the same time. Two routes are available, one via X and one via Y (Refer to the left panel of following Figure). The roads from A to X, and from Y to B are both short and narrow; in each case, one car takes 6 minutes, and each additional car increases the travel time per car by 3 minutes. (If two cars drive from A to X, for example, each car takes 9 minutes.) The roads from A to Y, and from X to B are long and wide; on A to Y one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute; on X to B one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minutes. Formulate this situation as a strategic game



Original network. Network with new road from X to Y.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 279 Question Id : 6406531290053 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Choose the correct alternative

Options :

6406534343477. ✔ In Nash equilibrium two people take each route

6406534343478. ✖ For the NE action profile, each person’s travel time is 28 minutes

6406534343479. ✖ Both In Nash equilibrium two people take each route and For the NE action profile, each person’s travel time is 28 minutes

6406534343480. ✖ None

Question Number : 280 Question Id : 6406531290054 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Now suppose that a relatively short, wide road is built from X to Y, giving each person four options for travel from A to B: A–X–B, A–Y–B, A–X–Y–B, and A–Y– X–B (As shown in the right panel of the given figure). Assume that a person who takes A–X–Y–B travels the A–X portion at the same time as someone who takes A–X–B, and the Y–B portion at the same time as someone who takes A–Y–B. (Think of there being constant flows of traffic.) On the road between X and Y, one car takes 7 minutes and each additional car increases the travel time per car by 1 minute. Find the Nash equilibrium in this new situation

Choose the correct alternative

Options :

6406534343481. ✖ In any Nash equilibrium, one person takes A–X–B, two people take A–X–Y–B, and one person takes A–Y–B

6406534343482. ✖ In this equilibrium profile, every person’s travel time increases compared to the case when new road was not there

6406534343483. ✔ Both In any Nash equilibrium, one person takes A–X–B, two people take A–X–Y–B, and one person takes A–Y–B and In this equilibrium profile, every person’s travel time increases compared to the case when new road was not there

6406534343484. ✖ None

Sub-Section Number :	3
Sub-Section Id :	640653204543
Question Shuffling Allowed :	No

Question Id : 6406531290055 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (281 to 287)

Question Label : Comprehension

Determined to crack down on the drug trade, a city mayor puts more officers out on patrol to disrupt the business of drug dealers. A drug dealer in a neighborhood can work his trade either on a street corner or in the park. Each day, he decides where to set up shop, knowing that word about his location will travel among users. Because a good snitch is lacking, word does not travel to the police. The police officer on the beat then needs to decide whether she will patrol the park or the street corner, while not knowing where the drug dealer is hanging out that day.
The decision of the officer and the dealer determine the extent of drug trades that day. Let suppose the total number of potential trades in the market is 500. A dealer's payoff is determined by the number of trades he consummates, while the officer's payoff is based on the number of trades she disrupts. The table below illustrates the payoffs for both parties according to their respective strategies in the game.

		Drug Dealer	
		Street Corner	Park
Police Officer	Street Corner	400, 100	0, 500
	Park	100, 400	300, 200

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 281 Question Id : 6406531290056 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct statement

Options :

6406534343485. ✖ There is a unique Nash equilibrium in pure strategy

6406534343486. ✖ There are finitely many Nash equilibria in this game in pure strategy

6406534343487. ✔ There is no Nash equilibrium in this game in pure strategy

6406534343488. ✖ None

Question Number : 282 Question Id : 6406531290057 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Choose the correct statement .

Options :

6406534343489. ✔ There is a unique Nash equilibrium in this game

6406534343490. ✖ There are finitely many Nash equilibria in this game

6406534343491. ✖ There is no Nash equilibrium in this game

6406534343492. ✖ None

Question Number : 283 Question Id : 6406531290058 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

If randomization probability of a police officer for patrolling the streets is p/q, then p =___________

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 284 Question Id : 6406531290059 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

If randomization probability of a police officer for patrolling the streets is p/q, then q =_____________

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 285 Question Id : 6406531290060 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

If randomization probability of a police officer for patrolling the park is r/s, then r =____________

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 286 Question Id : 6406531290061 Question Type : SA

Correct Marks : 0.5

Question Label : Short Answer Question

If randomization probability of a police officer for patrolling the park is r/s, then s =______________

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Question Number : 287 Question Id : 6406531290062 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

If randomization probability of the drug dealer choosing street is β, then β= ______________

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.5

Sub-Section Number :	4
Sub-Section Id :	640653204544
Question Shuffling Allowed :	No

Question Id : 6406531290063 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (288 to 295)

Question Label : Comprehension

Consider a two-player game in which player 1 can choose A or B. The game ends if he chooses A while it continues to player 2 if he chooses B. Player 2 can either choose C or D, with the game ending after C and continuing again with player 1 after D. Player 1 then can choose E or F, and then the game ends after each of these choices.
Imagine that the payoffs following choice A by player 1 are (2, 0), following C by player 2 are (3, 1), following E by player 1 are (0, 0) and following F by player 1 are (1, 2) Model this as an extensive form game and answer the given subquestions:

Sub questions

Question Number : 288 Question Id : 6406531290064 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

This is a game of

Options :

6406534343498. ✔ Perfect information

6406534343499. ✖ Imperfect information

6406534343500. ✖ None

6406534343501. ✖ Can't be determined

Question Number : 289 Question Id : 6406531290065 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many terminal nodes does this game have?

Options :

6406534343502. ✖ 3

6406534343503. ✖ 2

6406534343504. ✔ 4

6406534343505. ✖ 5

Question Number : 290 Question Id : 6406531290066 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

How many information sets does this game have?

Options :

6406534343506. ✖ 2

6406534343507. ✔ 3

6406534343508. ✖ 4

6406534343509. ✖ 1

Question Number : 291 Question Id : 6406531290067 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
How many pure strategies does player 1 have?
Options :
6406534343510. ✖ 1
6406534343511. ✖ 2
6406534343512. ✖ 3
6406534343513. ✔ 4

Question Number : 292 Question Id : 6406531290068 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
How many pure strategies does player 2 have?
Options :
6406534343514. ✖ 1
6406534343515. ✔ 2
6406534343516. ✖ 3
6406534343517. ✖ 4

Question Number : 293 Question Id : 6406531290069 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
How many pure strategy Nash equilibria does this game have?
Options :
6406534343518. ✖ 1
6406534343519. ✖ 2
6406534343520. ✔ 3
6406534343521. ✖ 4

Question Number : 294 Question Id : 6406531290070 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
Which one of the Nash equilibria is also a subgame perfect equilibrium?
Options :
6406534343522. ✖ (AE, D)
6406534343523. ✔ (AF, D)
6406534343524. ✖ (BE, C)
6406534343525. ✖ (BF, C)

Question Number : 295 Question Id : 6406531290071 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

No. of sub games this game has

Options :

6406534343526. ✖ 1

6406534343527. ✖ 2

6406534343528. ✔ 3

6406534343529. ✖ 4

Sub-Section Number :	5
Sub-Section Id :	640653204545
Question Shuffling Allowed :	No